

BRAINPOWER EDUCATION

Specialist Mathematics Units 1/2 - ComplexNumbers2026

Assessment Guide / Marking Scheme - 30 marks

IMPORTANT: Question 1 is mathematically invalid as printed. The treatment below prevents candidates being penalised.

Q1 - INVALID AS PRINTED - 7

The printed polynomial is $z^3 + (2-2i)z^2 - 4iz - 8 = 0$. Substituting $z=2i$ gives -8, not 0. Therefore part (a) is false and parts (b)-(c) cannot be completed as intended. Recommended: award 7/7 to all candidates or exclude Q1 from the denominator. A likely intended correction is changing the constant -8 to 0; then the roots are $2i$, 0 , -2 and the triangle area is 2.

Mark	Evidence required
7	Do not penalise candidates for the defective item.

Q2a - 2

$w = -8\sqrt{2} + 8\sqrt{2}i = 16 \operatorname{cis}(3\pi/4)$.

Mark	Evidence required
1	Modulus 16.
1	Argument $3\pi/4$.

Q2b - 3

Cube roots: $16^{1/3} \operatorname{cis}(\pi/4)$, $16^{1/3} \operatorname{cis}(11\pi/12)$, $16^{1/3} \operatorname{cis}(19\pi/12)$.

Mark	Evidence required
1	Correct root modulus.
1	Correct general argument.
1	All three roots.

Q2c - 4

Smallest positive argument root is $16^{1/3} \operatorname{cis}(\pi/4)$. Its fourth power is $16^{4/3} \operatorname{cis}(\pi) = -32 \operatorname{cube_root}(2)$.

Mark	Evidence required
1	Selects correct root.
1	Raises modulus/argument.
1	Uses $\operatorname{cis}(\pi) = -1$.
1	Exact Cartesian result $-32 \operatorname{cube_root}(2) + 0i$.

Q3a - 3

$D = u^{15}/u^3 = u^{12} = \operatorname{cis}(4\pi/3) = -1/2 - (\sqrt{3}/2)i$.

Mark	Evidence required
1	u^{12} .
1	Argument $4\pi/3$.
1	Correct Cartesian form.

Q3b - 3

$1 + \operatorname{cis}(a) = 2\cos(a/2) \operatorname{cis}(a/2)$. Raising to n gives $(1 + \operatorname{cis}(a))^n = (2\cos(a/2))^n \operatorname{cis}(na/2)$.

Mark	Evidence required
1	Uses half-angle identities.
1	Obtains base identity.
1	Applies De Moivre.

Q3c - 2

$$(1 + \text{cis}(\pi/3))^6 = (\sqrt{3})^6 \text{cis}(\pi) = -27.$$

Mark	Evidence required
1	Correct substitution.
1	$-27 + 0i$.

Q4a - 2

$$z_k = \text{cis}((2k+1)\pi/5), \quad k=0,1,2,3,4.$$

Mark	Evidence required
1	Correct general fifth-root arguments.
1	All five roots.

Q4b - 2

For $z^5 + 1$, the z^4 coefficient is zero, so by Vieta the sum of all five roots is 0.

Mark	Evidence required
1	Valid Vieta/roots-of-unity reasoning.
1	Sum = 0.

Q4c - 2

$$z^5 + 1 = \text{product}(z - w_k). \quad \text{At } z=2, \text{ product}(2 - w_k) = 2^5 + 1 = 33.$$

Mark	Evidence required
1	Evaluates polynomial at $z=2$.
1	33.